A monotonicity-free multidimensional Gamma-liminf and matching recovery for ridge targets

Candidate partial result for arXiv:1909.02162

11 August 2026

Status: candidate partial result, likely valid. The full higher-dimensional Gamma-limsup remains open. The result below removes monotonicity from the complete Gamma-liminf and proves the matching Gamma-limsup for all one-coordinate ridge targets on cylindrical domains.

1 Source question

Brezis and Nguyen prove in one dimension that a broad class of nonlocal, nonconvex energies Gamma-converges without assuming that the interaction law is nondecreasing [1]. On PDF page 3 they ask whether that monotonicity hypothesis can also be removed when $d > 1$.

Let $p \geq 1$, let $\Omega \subset \mathbb{R}^d$ be open, and let $\varphi : [0, \infty) \to [0, \infty)$ have the regularity assumed in [1]: it is defined everywhere, continuous except at finitely many positive points where both one-sided limits exist, and $\varphi(0) = 0$. Assume, for some $\alpha, \beta > 0$,

$$\varphi(t) \leq \alpha t^{p+1} \quad (0 \leq t \leq 1), \qquad \varphi(t) \leq \beta \quad (t \geq 0), \qquad \int_0^\infty \varphi(t) t^{-p-1}\, dt = \frac{1}{2}. \tag{1}$$

No monotonicity is imposed. Put

$$\mathcal{F}_\delta^d(u;\Omega) := \delta^p \int_\Omega \int_\Omega \frac{\varphi(|u(x) - u(y)|/\delta)}{|x-y|^{d+p}}\, dx\, dy. \tag{2}$$

Let $\kappa = \kappa(\varphi, p) \in [0, 1]$ be the one-dimensional Gamma-limit constant of [1, Theorem 1.1], with the normalization in (1), and define

$$\gamma_{d,p} := \int_{\mathbb{S}^{d-1}} |\sigma \cdot e|^p\, d\sigma, \qquad c_{d,p} := \frac{\gamma_{d,p}}{2}, \tag{3}$$

where e is any unit vector.

For $p > 1$, write

$$\mathcal{E}_p(u;\Omega) = \int_\Omega |\nabla u|^p\, dx \quad \text{if } u \in W^{1,p}(\Omega),$$

and $\mathcal{E}_p = +\infty$ otherwise. For $p = 1$, set $\mathcal{E}_1(u;\Omega) = |Du|(\Omega)$ for $u \in BV(\Omega)$ and $+\infty$ otherwise. As in [1], when $\kappa = 0$ the product $\kappa\mathcal{E}_p$ means the zero functional on all of L^p.

$$\lim_{\delta\to 0} \Lambda_\delta(u, I) = \int_I |u'|^p \, dx. \tag{1.7}$$

The conclusion of Proposition 1.1 under the assumption *i*) follows from [6, Theorem 1] (the only remaining case to be considered is the case $I = (0, +\infty)$ which can be deduced from the cases I bounded and $I = \mathbb{R}$ by standard arguments). The proof of Proposition 1.1 under the assumption *ii*) appeared in [5, proof of Proposition 1] under the additional assumption

$$\varphi \text{ is non-decreasing;} \tag{1.8}$$

however, this assumption can be easily removed from the proof. The conclusion of Proposition 1.1 contrasts with the conclusion of Theorem 1.1 since it may happen, for some functions φ $\big($e.g. $\varphi := \frac{p}{2}\mathbb{1}_{(1,+\infty)}\big)$, that κ is *strictly* less than 1 (see [8]); an explicit value of κ for this φ is given in [1]. As established in [2], it may happen that $\kappa(\varphi) = 1$ for some φ.

This work is a follow-up of our previous papers [5,6] where we investigated a similar problem in any dimension $d \geq 1$. More precisely, I is replaced by a domain $\Omega \subset \mathbb{R}^d$ and the RHS in (1.1) is replaced by

$$\int_\Omega \int_\Omega \frac{\varphi(|u(x) - u(y)|)}{|x-y|^{p+d}} \, dx \, dy.$$

Assuming (1.2), (1.3), and the *additional condition* (1.8), we established in [5,6] the Γ-convergence of Λ_δ to a multiple of $\int_\Omega |\nabla u|^p \, dx$. In these works, the monotonicity assumption (1.8) played a crucial role at almost every level of the proofs. The proof of Theorem 1.1 has its roots in [3,9,5,6]. However, many new ideas are required to overcome the lack of assumption (1.8). We do not know whether (1.8) can be removed when $d > 1$.

2. Proof of the main result

Figure 1: The source question on PDF page 3 of [1].

2 Idea of the proof

The d-dimensional energy is exactly one half of the average of the one-dimensional energies on all oriented lines. This identity makes the one-dimensional Gamma-liminf available on almost every slice; Fatou's lemma then turns the averaged directional Sobolev or BV seminorm into $\gamma_{d,p}\mathcal{E}_p$. This proves the lower half of the desired Gamma-limit without monotonicity.

For recovery, a general multidimensional gluing argument is still missing. There is nevertheless an exact nontrivial class where no gluing is needed. If $\Omega = Ie + D$ is a cylinder and $u(te+z) = g(t)$, lift a one-dimensional recovery family for g constantly across D. Integrating the Riesz kernel over the transverse variables bounds it by $c_{d,p}|t-s|^{-p-1}$, and the identity $c_{d,p} = \gamma_{d,p}/2$ gives exactly the same coefficient as the global lower bound.

3 The partial theorem

Theorem 1 (Global lower bound and exact ridge recovery). *Assume* (1) *and let* $d \geq 2$.

(i) *For every open* $\Omega \subset \mathbb{R}^d$*, every sequence* $\delta_n \downarrow 0$*, and every* $u_n \to u$ *in* $L^p(\Omega)$*,*

$$\liminf_{n\to\infty} \mathcal{F}^d_{\delta_n}(u_n; \Omega) \geq c_{d,p} \, \kappa \, \mathcal{E}_p(u; \Omega). \tag{4}$$

(ii) *Let* $e \in \mathbb{S}^{d-1}$*, let* $I \subset \mathbb{R}$ *be an open interval, and let* $D \subset e^\perp$ *be open with finite measure. Set* $\Omega = \{te + z : t \in I,\ z \in D\}$*. For every* $g \in L^p(I)$ *and* $u(te+z) = g(t)$*, there is* $u_\delta \to u$ *in* $L^p(\Omega)$ *such that*

$$\limsup_{\delta \downarrow 0} \mathcal{F}_\delta^d(u_\delta; \Omega) \le c_{d,p}\, \kappa\, \mathcal{E}_p(u; \Omega). \tag{5}$$

Consequently the lower bound in (4) *is sharp at every such ridge target, even against arbitrary (not necessarily ridge) perturbing sequences.*

4 Proof

4.1 An exact line decomposition

Fix $\sigma \in \mathbb{S}^{d-1}$ and $z \in \sigma^\perp$, and set

$$\Omega_{\sigma,z} := \{t \in \mathbb{R} : z + t\sigma \in \Omega\}, \qquad u_{\sigma,z}(t) := u(z + t\sigma).$$

Denote by $\mathcal{F}_\delta^1(\,cdot\,; J)$ the one-dimensional functional from [1]. Polar coordinates $y = x + h\sigma$, followed by the foliation $x = z + t\sigma$, give the exact identity

$$\mathcal{F}_\delta^d(u; \Omega) = \frac{1}{2} \int_{\mathbb{S}^{d-1}} \int_{\sigma^\perp} \mathcal{F}_\delta^1(u_{\sigma,z}; \Omega_{\sigma,z})\, dz\, d\sigma. \tag{6}$$

Indeed, the Jacobian h^{d-1} changes $|x-y|^{-d-p}$ into h^{-p-1}, and the factor $1/2$ appears because the full one-dimensional double integral counts both orientations of every unordered pair.

4.2 Proof of the lower bound

It is enough to consider a subsequence along which the left side of (4) is a limit. Passing to a further subsequence, we may assume that sliced L^p convergence holds for almost every (σ, z). To see this, use Fubini and

$$\int_{\mathbb{S}^{d-1}} \int_{\sigma^\perp} \|(u_n - u)_{\sigma,z}\|^p_{L^p(\Omega_{\sigma,z})}\, dz\, d\sigma = |\mathbb{S}^{d-1}|\, \|u_n - u\|^p_{L^p(\Omega)}, \tag{7}$$

then choose a summable subsequence.

For almost every (σ, z), the open set $\Omega_{\sigma,z}$ is a countable union of disjoint intervals. On every finite collection of those components, nonnegativity lets us discard all other and all cross-component interactions. The one-dimensional Gamma-liminf theorem [1, Theorem 1.1], followed by monotone convergence over the components, yields

$$\liminf_{n\to\infty} \mathcal{F}_{\delta_n}^1((u_n)_{\sigma,z}; \Omega_{\sigma,z}) \ge \kappa\, \mathcal{E}_p^1(u_{\sigma,z}; \Omega_{\sigma,z}). \tag{8}$$

Fatou's lemma in (6) now gives

$$\liminf_{n} \mathcal{F}_{\delta_n}^d(u_n; \Omega) \ge \frac{\kappa}{2} \int_{\mathbb{S}^{d-1}} \int_{\sigma^\perp} \mathcal{E}_p^1(u_{\sigma,z}; \Omega_{\sigma,z})\, dz\, d\sigma. \tag{9}$$

For $p > 1$, Sobolev slicing and rotation invariance give

$$\begin{aligned}\int_{\mathbb{S}^{d-1}} \int_{\sigma^\perp} \mathcal{E}_p^1(u_{\sigma,z}; \Omega_{\sigma,z})\, dz\, d\sigma &= \int_\Omega \int_{\mathbb{S}^{d-1}} |\nabla u(x) \cdot \sigma|^p\, d\sigma\, dx \\ &= \gamma_{d,p} \int_\Omega |\nabla u|^p\, dx.\end{aligned}$$

The standard slicing characterization also shows that finite right side in (9), when $\kappa > 0$, forces $u \in W^{1,p}(\Omega)$.

For $p = 1$, the BV slicing theorem and the polar decomposition $Du = \nu_u |Du|$ give [3, Section 3.11]

$$\begin{aligned}\int_{\mathbb{S}^{d-1}} \int_{\sigma^\perp} |D(u_{\sigma,z})|(\Omega_{\sigma,z})\, dz\, d\sigma &= \int_{\mathbb{S}^{d-1}} |D_\sigma u|(\Omega)\, d\sigma \\ &= \int_\Omega \int_{\mathbb{S}^{d-1}} |\nu_u(x) \cdot \sigma|\, d\sigma\, d|Du|(x) \\ &= \gamma_{d,1} |Du|(\Omega).\end{aligned}$$

Again, finite averaged slice variation characterizes BV; if $\kappa = 0$, the source convention makes (4) the trivial nonnegative bound. This proves part (i).

4.3 Proof of ridge recovery

Rotate coordinates so that e is the first coordinate vector and write $x = (t, z) \in I \times D$. By the one-dimensional Gamma-limsup theorem there is $g_\delta \to g$ in $L^p(I)$ such that

$$\limsup_{\delta \downarrow 0} \mathcal{F}^1_\delta(g_\delta; I) \le \kappa\, \mathcal{E}^1_p(g; I). \tag{10}$$

Set $u_\delta(t, z) = g_\delta(t)$. Then $u_\delta \to u$ in $L^p(I \times D)$, and Tonelli's theorem gives

$$\mathcal{F}^d_\delta(u_\delta; I \times D) = \int_I \int_I \varphi_\delta(|g_\delta(t) - g_\delta(s)|) K_D(t - s)\, dt\, ds, \tag{11}$$

where $\varphi_\delta(r) = \delta^p \varphi(r/\delta)$ and

$$K_D(h) = \int_D \int_D \left(h^2 + |z - w|^2\right)^{-(d+p)/2} dz\, dw.$$

For $h \neq 0$, enlarge the inner transverse domain to all of $\mathbb{R}^{d-1}$:

$$\begin{aligned}K_D(h) &\le |D| \int_{\mathbb{R}^{d-1}} (h^2 + |q|^2)^{-(d+p)/2}\, dq \\ &= |D|\, \pi^{(d-1)/2} \frac{\Gamma((p+1)/2)}{\Gamma((d+p)/2)} |h|^{-p-1} = |D|\, c_{d,p} |h|^{-p-1}.\end{aligned} \tag{12}$$

The last equality follows either from the beta integral or from polar coordinates:

$$\gamma_{d,p} = 2\pi^{(d-1)/2} \frac{\Gamma((p+1)/2)}{\Gamma((d+p)/2)}.$$

Combining (11), (12), and (10),

$$\begin{aligned}\limsup_{\delta \downarrow 0} \mathcal{F}^d_\delta(u_\delta; I \times D) &\le |D| c_{d,p} \kappa \mathcal{E}^1_p(g; I) \\ &= c_{d,p} \kappa \mathcal{E}_p(u; I \times D).\end{aligned}$$

This is (5); part (i) supplies the matching lower bound against every multidimensional approximating family. $\square$

5 Verification and limitations

The calculation was audited in three independent ways: the polar-coordinate line formula fixes the factor 1/2; the beta integral computes the transverse kernel marginal; and the two constants agree through $c_{d,p} = \gamma_{d,p}/2$. The proof uses only nonnegativity and the source's one-dimensional theorem, never monotonicity of φ.

This packet does *not* solve the full source question. Its missing part is a Gamma-limsup for arbitrary multidimensional targets. A promising approach is to extend the source's repeated one-dimensional cell to a scalar map q_δ and try $u_\delta = q_\delta \circ u$. On affine regions this has the correct bulk mechanism, but the source cell is only controlled in L^p and may be discontinuous. Without monotonicity, there is presently no valid fundamental estimate controlling short-range interactions across interfaces where affine gradients change. Boundedness of φ alone produces a nonintegrable short-range majorant, so this point cannot be hidden inside a routine gluing assertion.

The bounded novelty check searched the run's exact-id/title indexes, its locally downloaded arXiv-source corpus through the 2026-06 build, and web queries using the exact question, authors, and nonmonotone interaction-law terminology. No later primary-source solution or duplicate of this partial theorem was located. Novelty confidence is therefore moderate, not exhaustive. Human review should concentrate on the measurable slice/subsequence step, the zero-κ convention, and the factor 1/2.

References

[1] H. Brezis and H.-M. Nguyen, *Γ-convergence of non-local, non-convex functionals in one dimension*, arXiv:1909.02162 (2019).

[2] H. Brezis and H.-M. Nguyen, *Non-local, non-convex functionals converging to Sobolev norms*, arXiv:1909.02160 (2019).

[3] L. Ambrosio, N. Fusco, and D. Pallara, *Functions of Bounded Variation and Free Discontinuity Problems*, Oxford Mathematical Monographs, 2000.